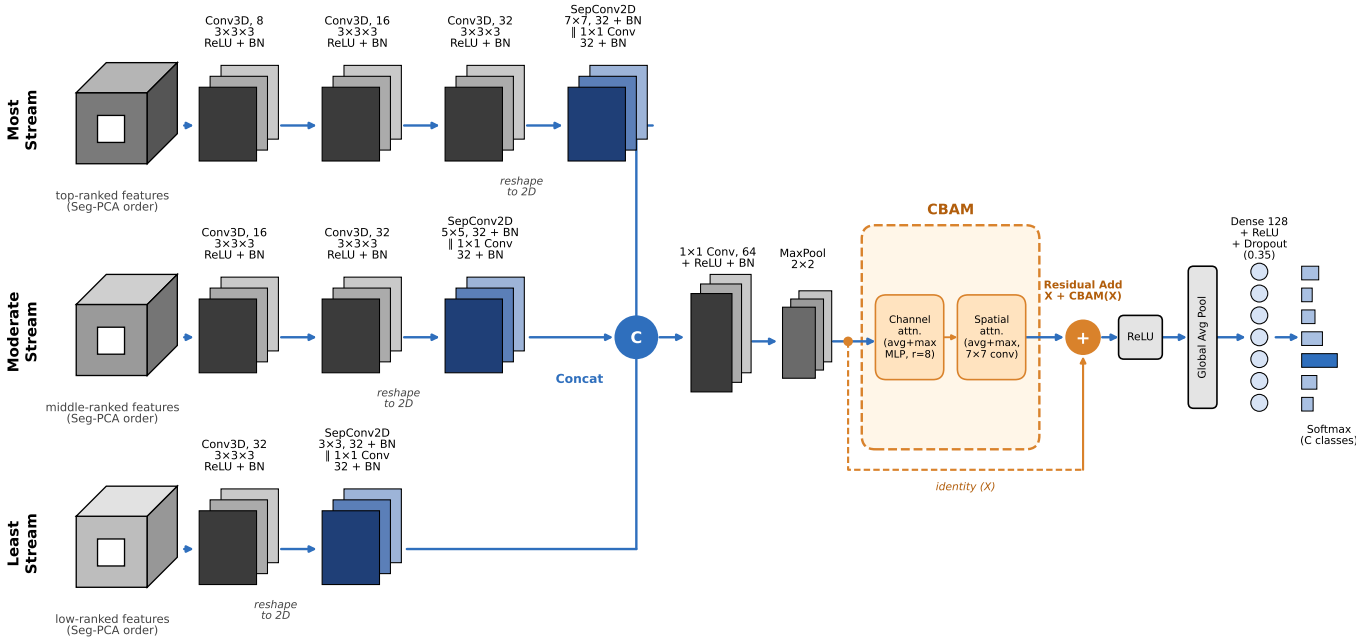
Most Stream
Conv3D, 8
3×3×3
ReLU + BN
Conv3D, 16
3×3×3
ReLU + BN
Conv3D, 32
3×3×3
ReLU + BN
SepConv2D
7×7, 32 + BN
‖ 1×1 Conv
32 + BN
top-ranked features
(Seg-PCA order)
reshape
to 2D
Moderate Stream
Conv3D, 16
3×3×3
ReLU + BN
Conv3D, 32
3×3×3
ReLU + BN
SepConv2D
5×5, 32 + BN
‖ 1×1 Conv
32 + BN
middle-ranked features
(Seg-PCA order)
reshape
to 2D
Least Stream
Conv3D, 32
3×3×3
ReLU + BN
SepConv2D
3×3, 32 + BN
‖ 1×1 Conv
32 + BN
low-ranked features
(Seg-PCA order)
reshape
to 2D
C
Concat
1×1 Conv, 64
+ ReLU + BN
MaxPool
2×2
CBAM
Channel
attn.
(avg+max
MLP, r=8)
Spatial
attn.
(avg+max,
7×7 conv)
Residual Add
X + CBAM(X)
+
ReLU
identity (X)
Global Avg Pool
Dense 128
+ ReLU
+ Dropout
(0.35)
Softmax
(C classes)